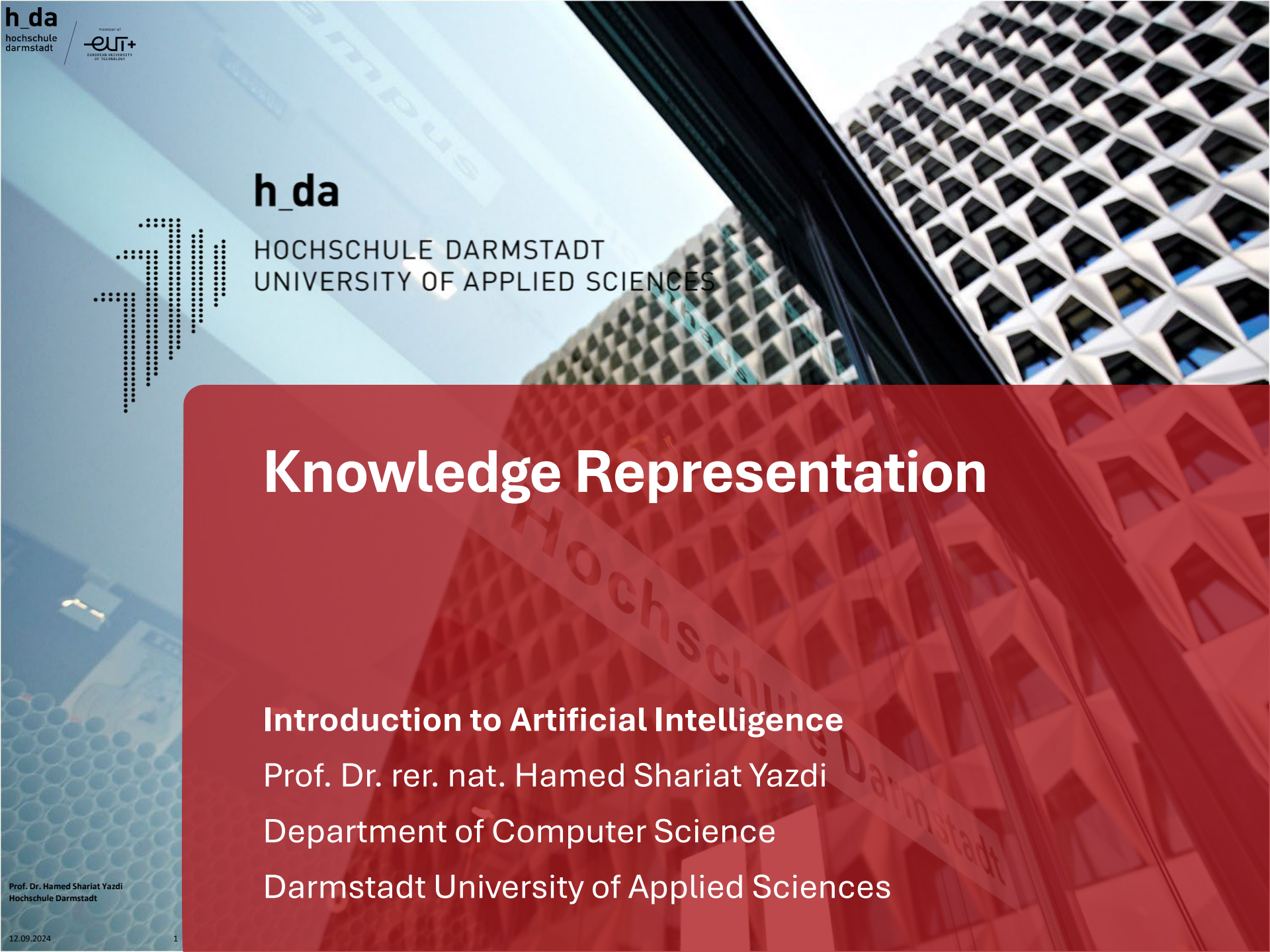


Knowledge Representation

Introduction to Artificial Intelligence

Prof. Dr. rer. nat. Hamed Shariat Yazdi

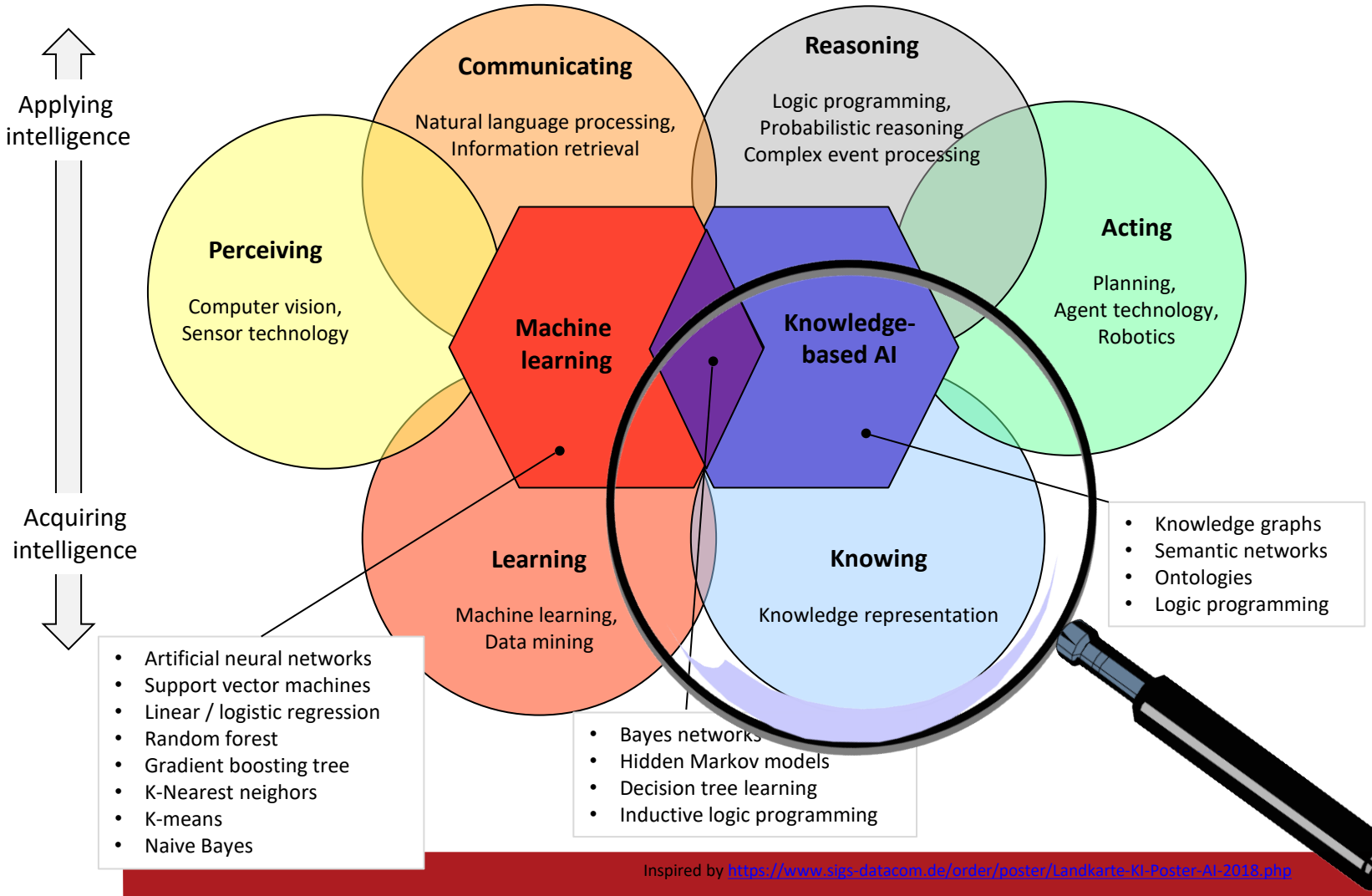
Department of Computer Science

Darmstadt University of Applied Sciences

Agenda

- Overview
- Knowledge Graphs
- RDF
- The TMDB Knowledge Graph
- SPARQL
- Apache Fuseki
- Using RDF and SPARQL in Python
- Quick check

AI landscape



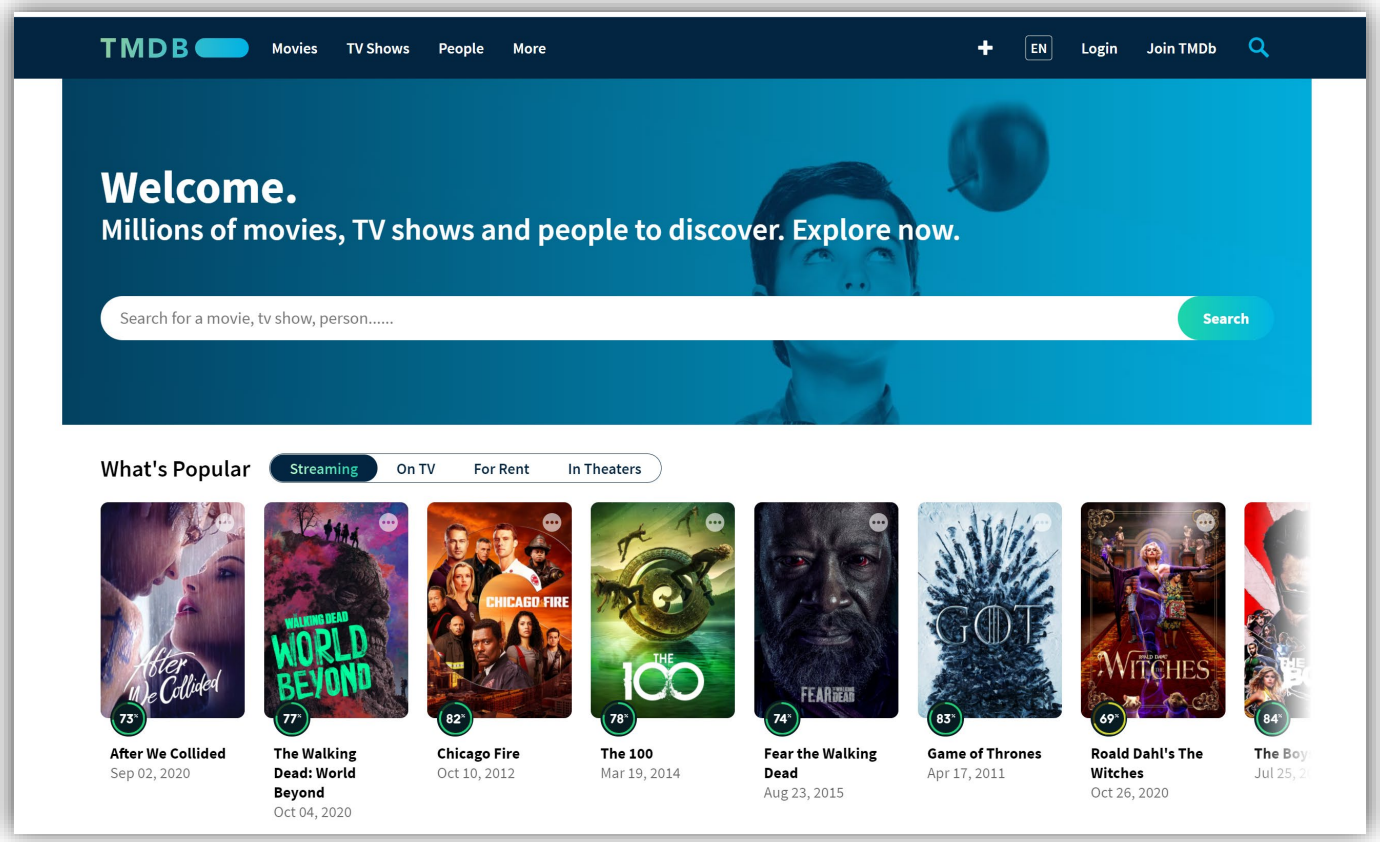
Inspired by https://www.sigs-datacom.de/order/poster/Landkarte-KI-Poster-AI-2018.php

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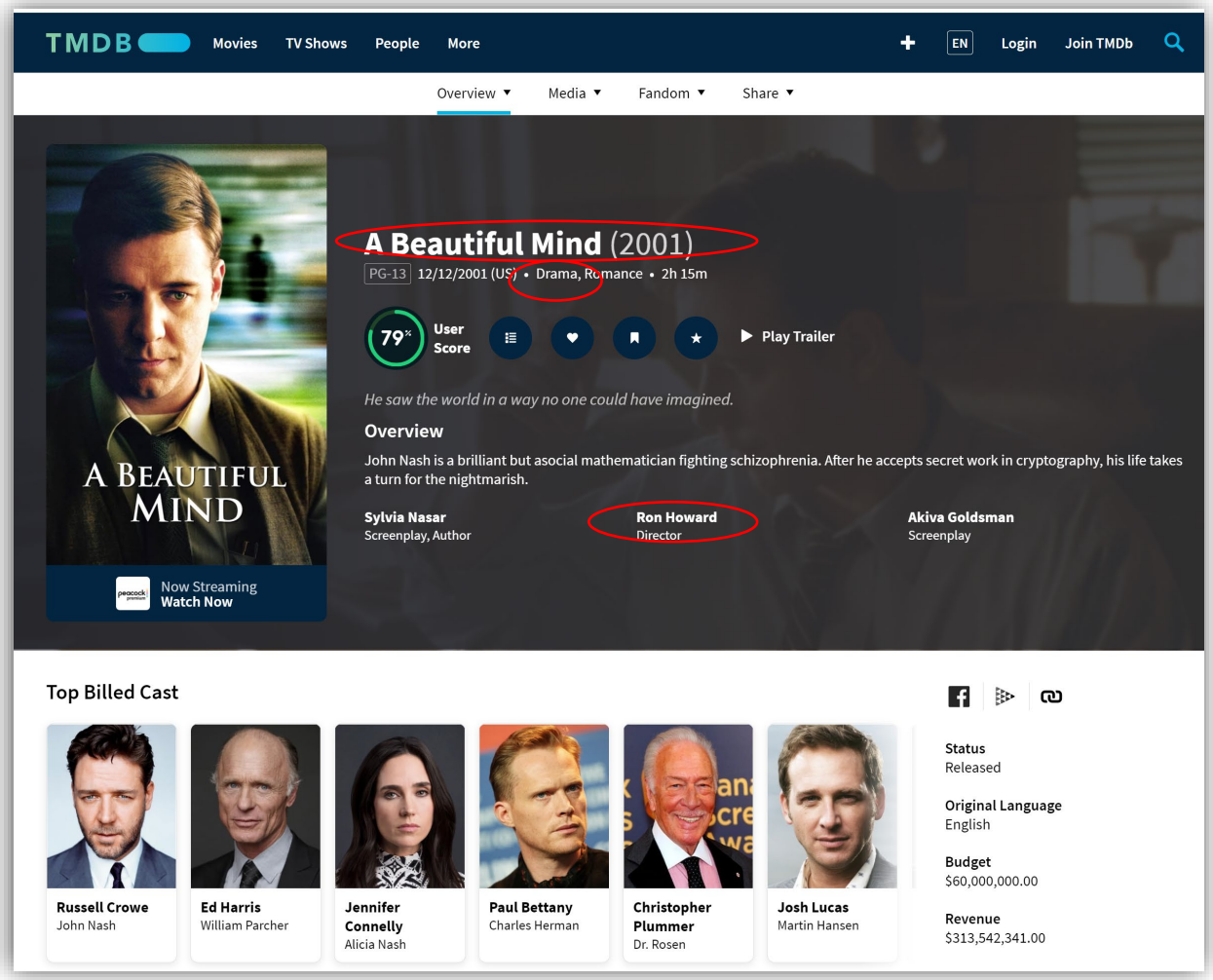
- Overview
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What do you know about movies?

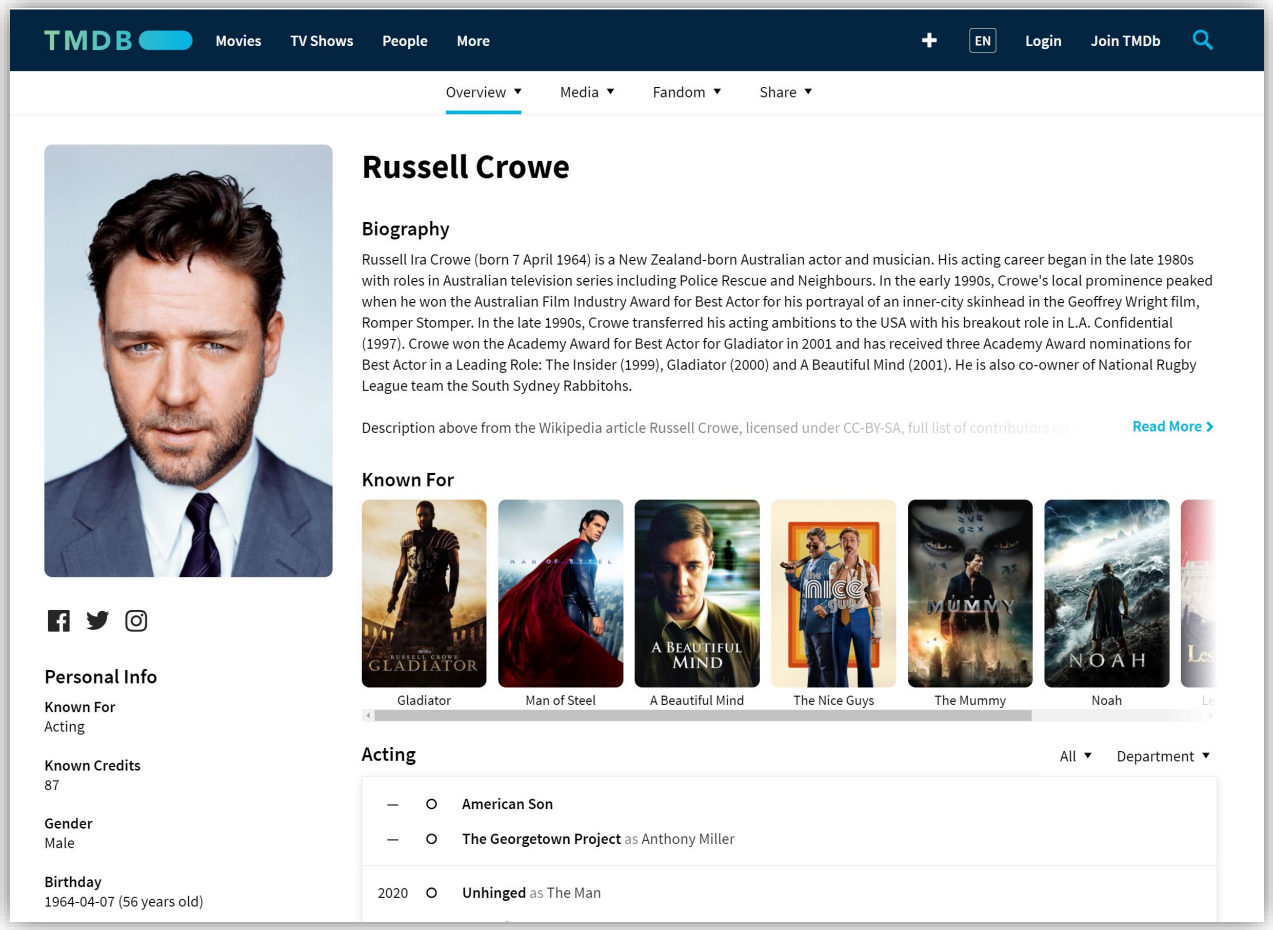


https://www.themoviedb.org

Knowledge about movies



https://www.themoviedb.org/movie/453



https://www.themoviedb.org/person/934

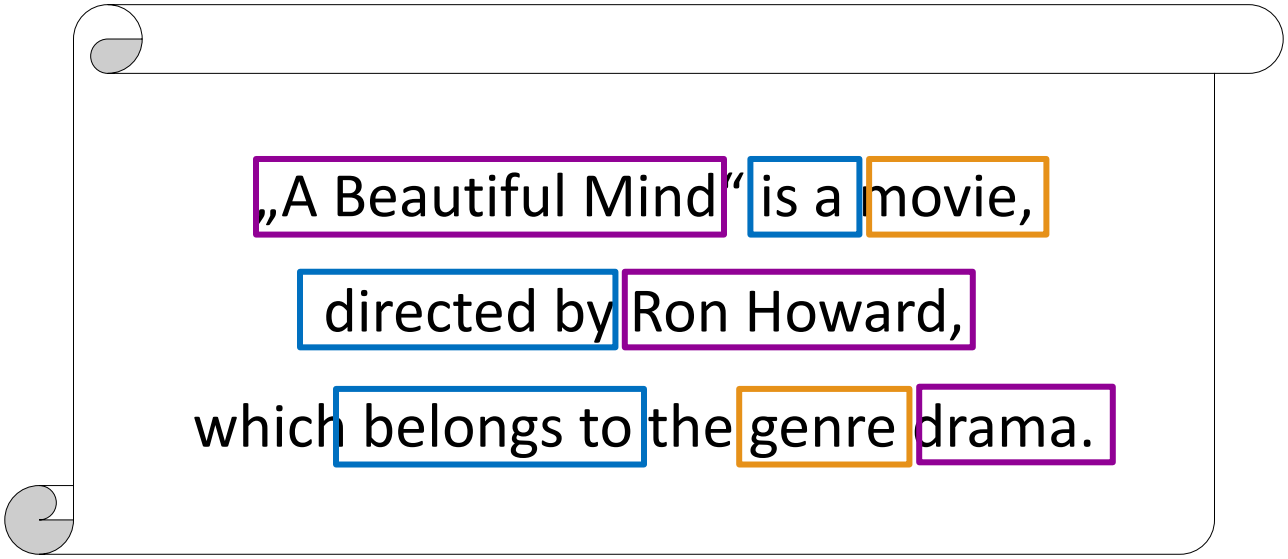
Knowledge about movies

„A Beautiful Mind“ is a movie,

directed by Ron Howard,

which belongs to the genre drama.

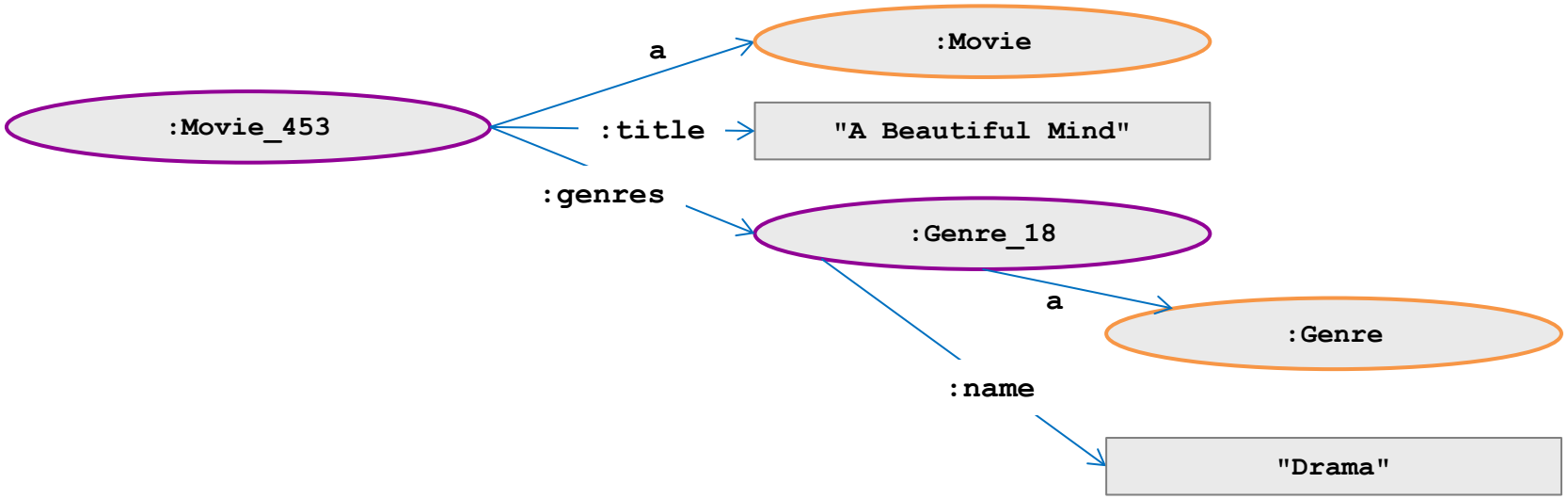
Representing knowledge about movies



Individuals | Classes | Relationships

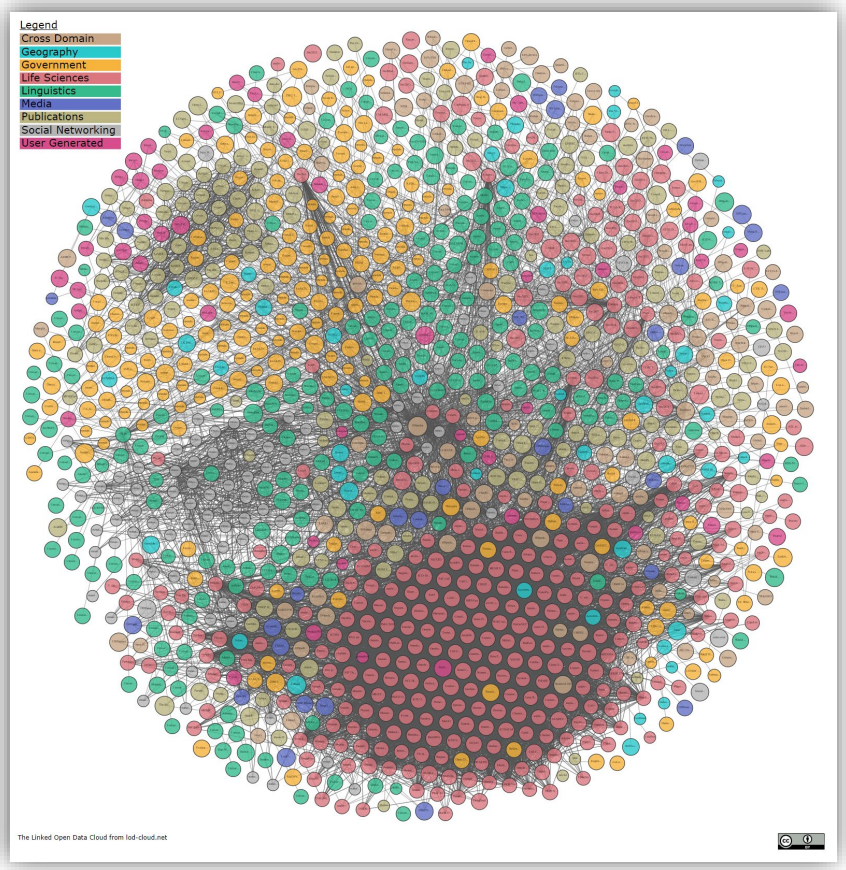
Knowledge Graph

- Related terms: Semantic net, linked data, ontology, thesaurus, …



Linked data:Thousands of knowledge graphs

- Large set of knowledge graphs
- Publicly available
- Tens of thousands entries each
- Standardized formats
- Numerous domains, e.g. life sciences, geography, media, social networking
- Examples: Wikidata, dbpedia, YAGO, MeSH, …



https://lod-cloud.net

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Resource Description Framework

https://www.w3.org/RDF

- **RDF**: Formal language for representing knowledge graphs
- Specified by **World Wide Web Consortium (W3C)** as part of the **Semantic Web** initiative
  https://www.w3.org/standards/semanticweb
- Notation used in this course: **Turtle**
  https://www.w3.org/TR/turtle

Resources in RDF are represented by URIs

Definition of a **base namespace**.
Allows abbreviating resources

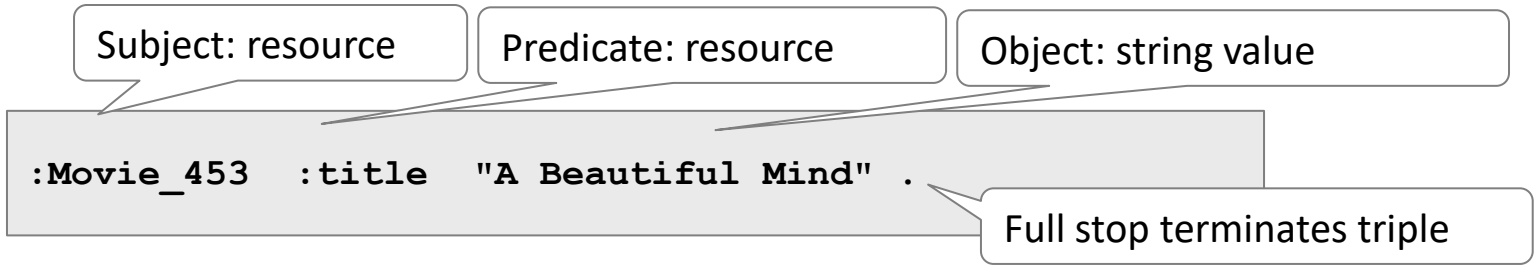
```
@base <https://www.themoviedb.org/kaggle-export/> .

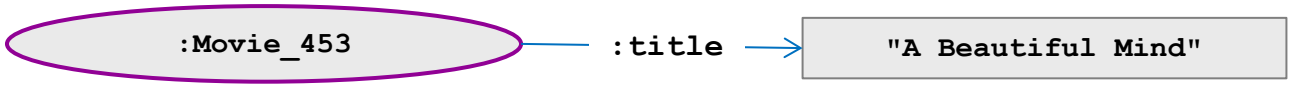
:Movie_453
```

A **resource** in RDF
Abbreviation for `<https://www.themoviedb.org/kaggle-export/Movie_453>`

Triples:
The basic statement notation of RDF

- RDF Triples include 3 parts:
  - **Subject**: resource
  - **Predicate**: resource
  - **Object**: resource or value (string, int, float, date, …)
- Example:





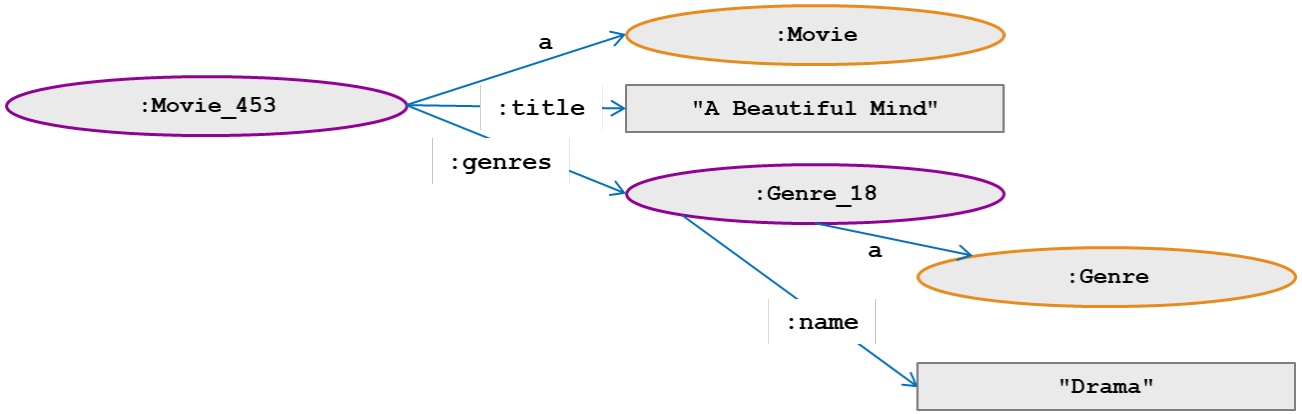
RDF triples represent knowledge graphs

```
:Movie_453 a :Movie ;
    :title  "A Beautiful Mind" ;
    :genres :Genre_18 .

:Genre_18 a :Genre ;
    :name "Drama" .
```

Semicolon chains several triples for the same subject

`a` means instance relationship (abbreviation for `rdf:type`)

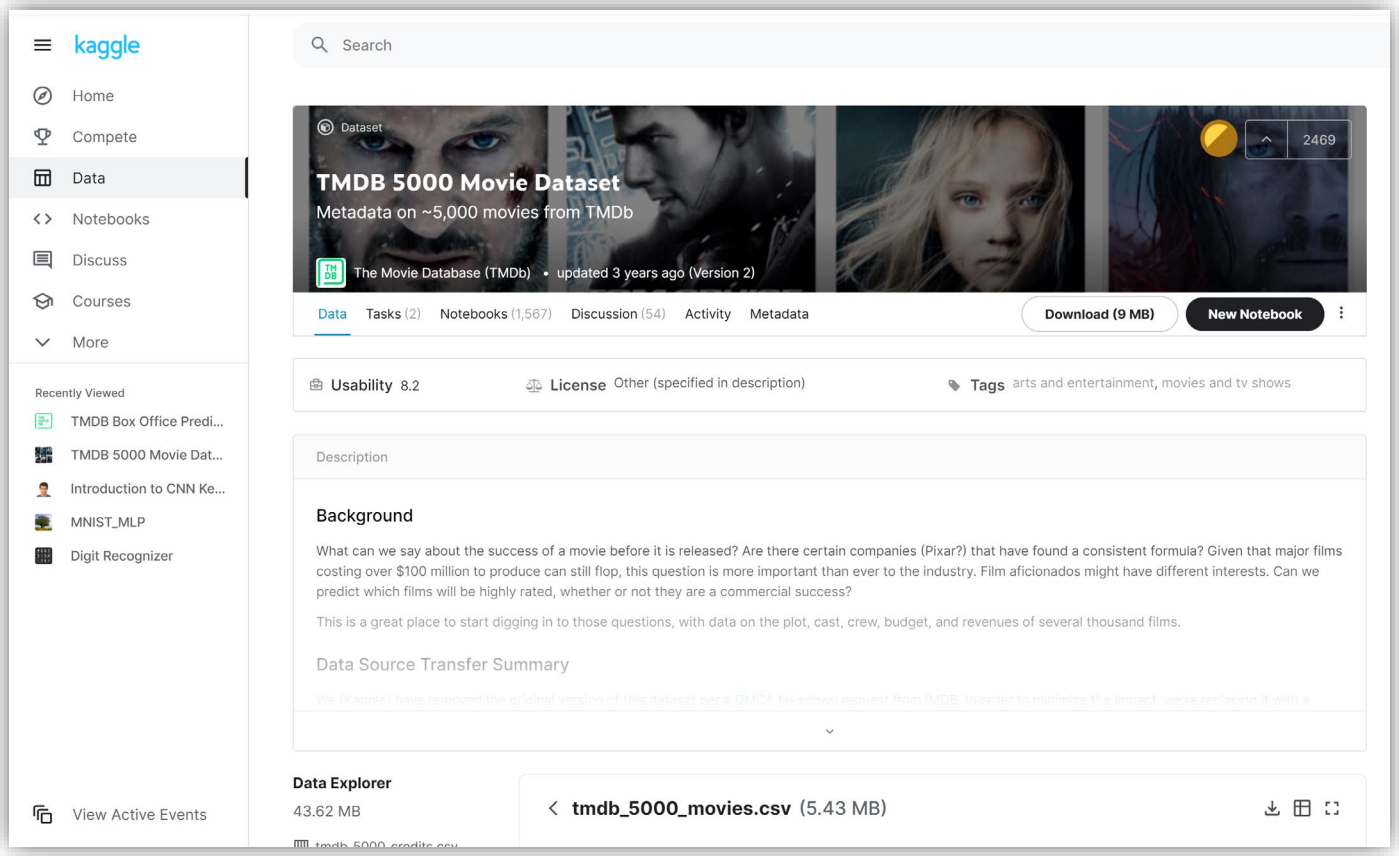


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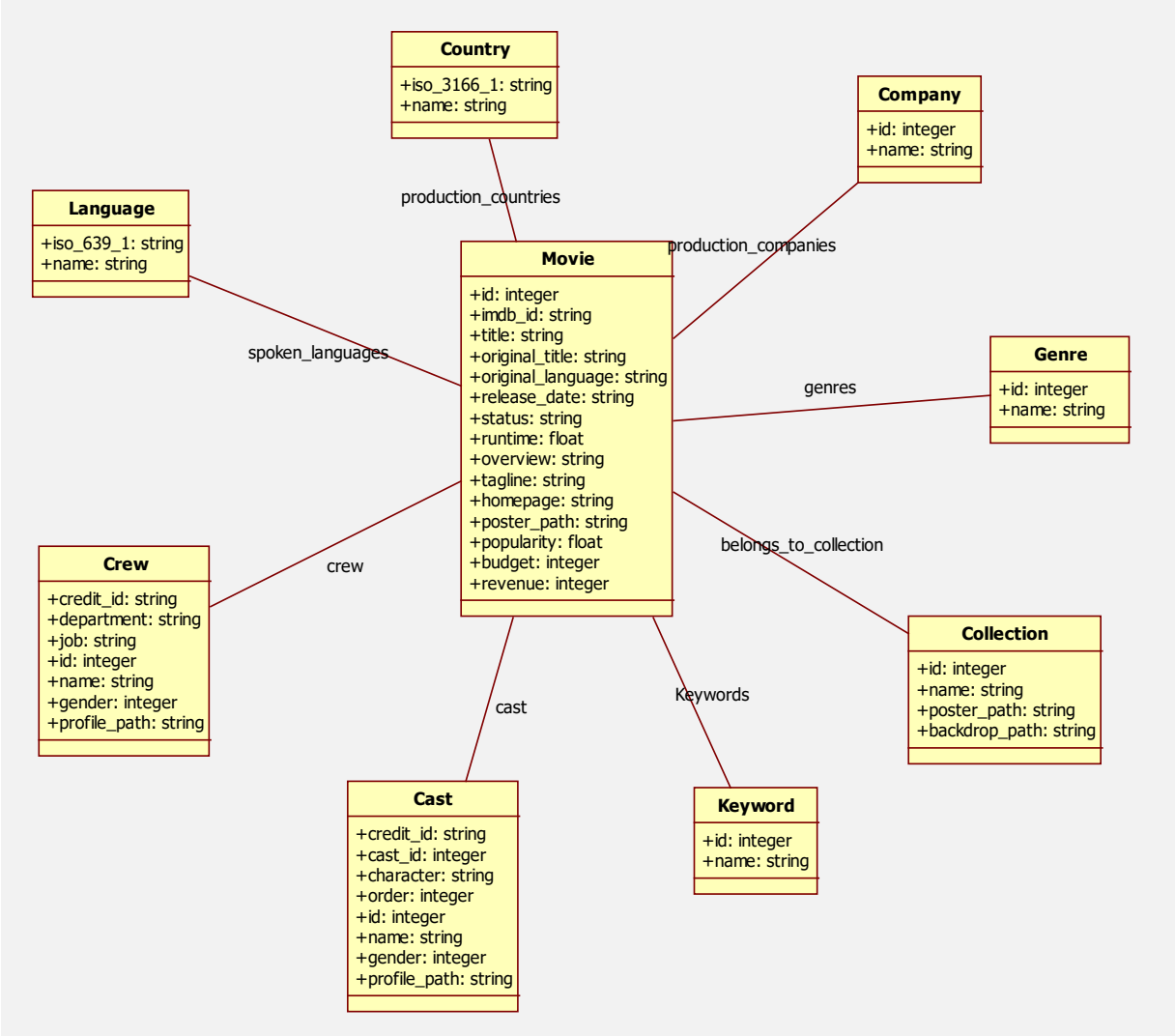
Dataset: Kaggle extract of TMDB

https://www.kaggle.com/c/tmdb-box-office-prediction/data?select=train.csv

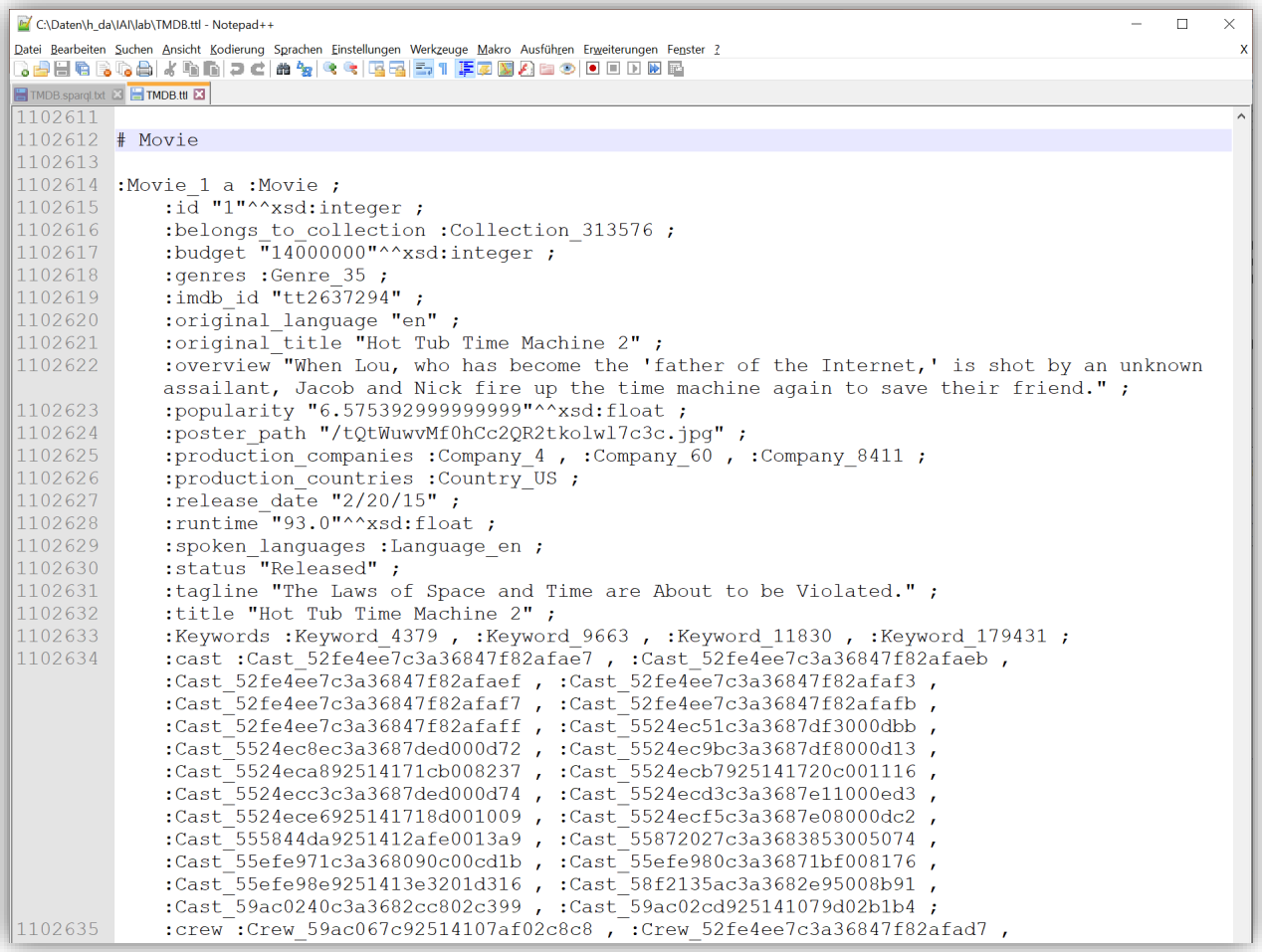


Data model

(UML class diagram)



RDF conversion: TMDB.ttl (via custom code for this course)

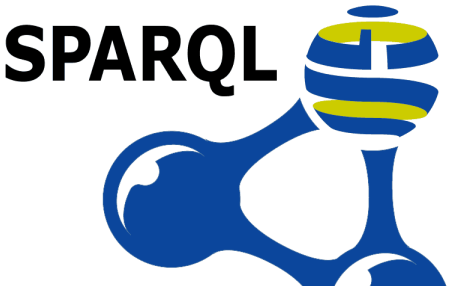



```
1102611
1102612 # Movie
1102613
1102614 :Movie_1 a :Movie ;
1102615     :id "1"^^xsd:integer ;
1102616     :belongs_to_collection :Collection_313576 ;
1102617     :budget "14000000"^^xsd:integer ;
1102618     :genres :Genre_35 ;
1102619     :imdb_id "tt2637294" ;
1102620     :original_language "en" ;
1102621     :original_title "Hot Tub Time Machine 2" ;
1102622     :overview "When Lou, who has become the 'father of the Internet,' is shot by an unknown
            assailant, Jacob and Nick fire up the time machine again to save their friend." ;
1102623     :popularity "6.575392999999999"^^xsd:float ;
1102624     :poster_path "/tQtWuwvMf0hCc2QR2tkolwl7c3c.jpg" ;
1102625     :production_companies :Company_4 , :Company_60 , :Company_8411 ;
1102626     :production_countries :Country_US ;
1102627     :release_date "2/20/15" ;
1102628     :runtime "93.0"^^xsd:float ;
1102629     :spoken_languages :Language_en ;
1102630     :status "Released" ;
1102631     :tagline "The Laws of Space and Time are About to be Violated." ;
1102632     :title "Hot Tub Time Machine 2" ;
1102633     :Keywords :Keyword_4379 , :Keyword_9663 , :Keyword_11830 , :Keyword_179431 ;
1102634     :cast :Cast_52fe4ee7c3a36847f82afae7 , :Cast_52fe4ee7c3a36847f82afaeb ,
            :Cast_52fe4ee7c3a36847f82afaef , :Cast_52fe4ee7c3a36847f82afaf3 ,
            :Cast_52fe4ee7c3a36847f82afaf7 , :Cast_52fe4ee7c3a36847f82afafb ,
            :Cast_52fe4ee7c3a36847f82afaff , :Cast_5524ec51c3a3687df3000dbb ,
            :Cast_5524ec8ec3a3687ded000d72 , :Cast_5524ec9bc3a3687df8000d13 ,
            :Cast_5524eca892514171cb008237 , :Cast_5524ecb7925141720c001116 ,
            :Cast_5524ecc3c3a3687ded000d74 , :Cast_5524ecd3c3a3687e11000ed3 ,
            :Cast_5524ece6925141718d001009 , :Cast_5524ecf5c3a3687e08000dc2 ,
            :Cast_555844da9251412afe0013a9 , :Cast_55872027c3a3683853005074 ,
            :Cast_55efe971c3a368090c00cd1b , :Cast_55efe980c3a36871bf008176 ,
            :Cast_55efe98e9251413e3201d316 , :Cast_58f2135ac3a3682e95008b91 ,
            :Cast_59ac0240c3a3682cc802c399 , :Cast_59ac02cd925141079d02b1b4 ;
1102635     :crew :Crew_59ac067c92514107af02c8c8 , :Crew_52fe4ee7c3a36847f82afad7 ,
```

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SPARQL:
Query Language for RDF



- Standardized by W3C
- Current Version: 1.1
- Specification: http://www.w3.org/TR/sparql11-query/

Simple SELECT query

Specifies the base namespace

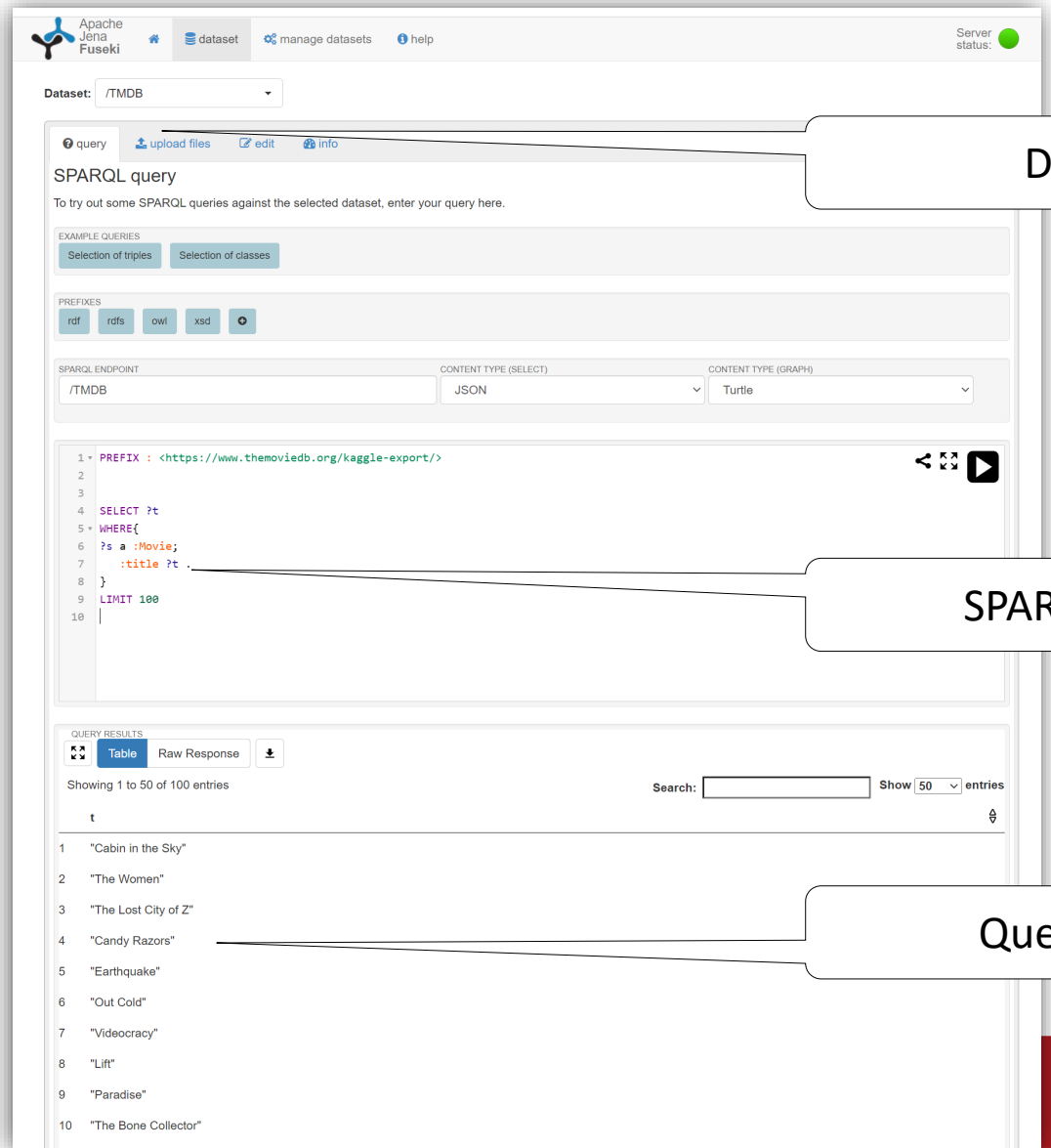
```
PREFIX : <https://www.themoviedb.org/kaggle-export/>

SELECT ?t
WHERE {
?m a :Movie ;
   :title ?t .
}
```

SELECT part: Query variable (starting with '?')

Query condition (WHERE part):
RDF triples with variables, specifying a pattern.
All variable assignments in the knowledge graph that exactly match this pattern will be returned as query result.

Query execution in Apache Fuseki



Dataset

SPARQL query

Query result

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SELECT *:
Selecting all query variables

Result table: One column for each query variable (`?s` and `?t`)

```
SELECT *
WHERE{
?s a :Movie;
   :title ?t .
}
```

QUERY RESULTS

Table | Raw Response

Showing 1 to 50 of 3,000 entries

Search: Show 50 entries

	s	t
1	:Movie_1622	"Cabin in the Sky"
2	:Movie_2191	"The Women"
3	:Movie_235	"The Lost City of Z"
4	:Movie_1187	"Candy Razors"
5	:Movie_2353	"Earthquake"
6	:Movie_861	"Out Cold"
7	:Movie_1420	"Videocracy"
8	:Movie_2687	"Lift"
9	:Movie_331	"Paradise"
10	:Movie_1006	"The Bone Collector"

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Multiple query variables

```
1  PREFIX : <https://www.themo
2
3  SELECT ?t ?o
4  WHERE{
5  ?s a :Movie;
6     :title ?t ;
7     :overview ?o .
8  }
9
10
```

Query variables

Result table: One column for each query variable

QUERY RESULTS

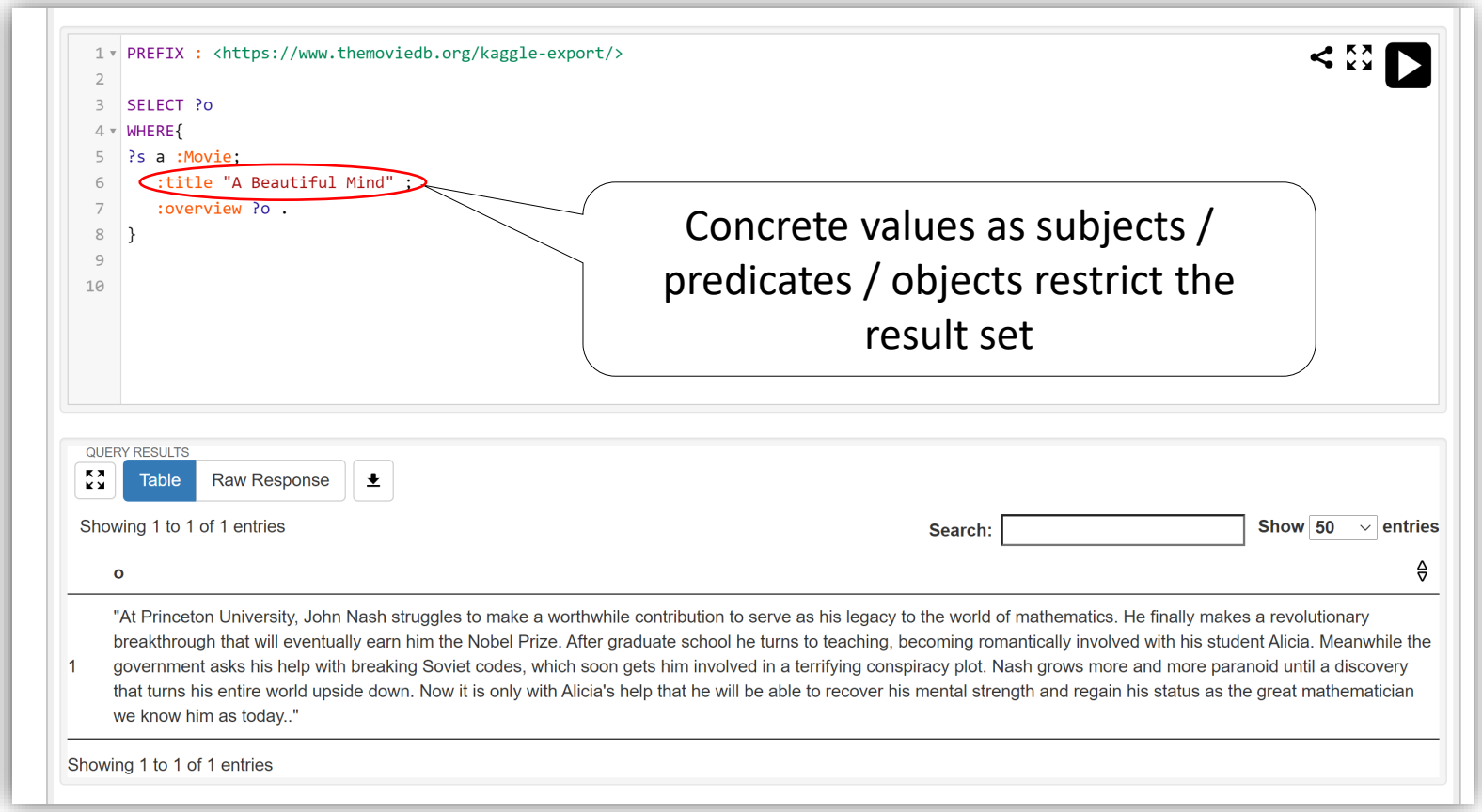
Table | Raw Response

Showing 1 to 50 of 2,992 entries

Search: Show 50 entries

	t	o
1	"Cabin in the Sky"	"Chronic gambler and carouser 'Little' Joe Jackson is shot by Domino Johnson at Jim Henry's gambling club over an outstanding gambling debt. Little Joe's wife, the God-fearing Petunia Jackson, prays not only for her husband's mortal life, but also his eternal soul as she's afraid that if he dies now, he, despite not being an evil man, won't make it into heaven. As Little Joe is close to death, he is visited by agents of both the Lord and of Lucifer. They make a deal with him: they will give him six months to atone for the errors of his human life. Once back on Earth, he won't remember the deal but both the Lord and Lucifer will be watching over him, trying to get him to see things their way. As both sides try to get Little Joe's soul, they figure that some of the most powerful tools they have at their disposal are the women in Little Joe's life: Petunia on behalf of the Lord, and Georgia Brown, a gold-digging floozy, on behalf of Lucifer."
2	"The Women"	"Wealthy Mary Haines is unaware her husband is having an affair with shopgirl Crystal Allen. Sylvia Fowler and Edith Potter discover this from a manicurist and arrange for Mary to hear the gossip. On the train taking her to a Reno divorce Mary meets the Countess and Miriam (in an affair with Fowler's husband). While they are at Lucy's dude ranch, Fowler arrives for her own divorce and the Countess meets fifth husband-to-be Buck. Back in New York, Mary's ex is now unhappily married to Crystal who is already in an affair with Buck."
3	"The Lost City of Z"	"At the dawn of the 20th century, British explorer Percy Fawcett journeys into the Amazon, where he discovers evidence of a previously unknown, advanced civilization that may have once inhabited the region. Despite being ridiculed by the scientific establishment, which views indigenous populations as savages, the determined Fawcett, supported by his devoted wife, son, and aide-de-camp, returns

Conditions in queries



Joins

```
1  PREFIX : <https://www.themoviedb.org/kaggle-export/>
2
3  SELECT ?m ?t ?g ?n
4  WHERE{
5  ?m a :Movie;
6     :title ?t ;
7     :genres ?g .
8  ?g :name ?n .
9  }
10
11
```

Using the **same variable** in various positions of the query condition enforces the **same value** each time

QUERY RESULTS

Table | Raw Response

Showing 1 to 50 of 7,511 entries

Search: Show 50 entries

	m	t	g	n
1	:Movie_1622	"Cabin in the Sky"	:Genre_14	"Fantasy"
2	:Movie_1622	"Cabin in the Sky"	:Genre_10402	"Music"
3	:Movie_2191	"The Women"	:Genre_18	"Drama"
4	:Movie_2191	"The Women"	:Genre_35	"Comedy"
5	:Movie_235	"The Lost City of Z"	:Genre_18	"Drama"
6	:Movie_235	"The Lost City of Z"	:Genre_36	"History"
7	:Movie_235	"The Lost City of Z"	:Genre_12	"Adventure"
8	:Movie_235	"The Lost City of Z"	:Genre_28	"Action"
9	:Movie_1187	"Candy Razors"	:Genre_80	"Crime"
10	:Movie_1187	"Candy Razors"	:Genre_18	"Drama"

DISTINCT queries

```
PREFIX : <https://www.themoviedb.org/kaggle-export/>

SELECT ?n
WHERE{
?m a :Movie;
   :title ?t ;
   :genres ?g .
?g :name ?n .
}
```

QUERY RESULTS

Table | Raw Response

Showing 1 to 50 of 7,511 entries

	n
1	"Fantasy"
2	"Music"
3	"Drama"
4	"Comedy"
5	"Drama"
6	"History"
7	"Adventure"
8	"Action"
9	"Crime"
10	"Drama"
11	"Thriller"

```
PREFIX : <https://www.themoviedb.org/kaggle-export/>

SELECT DISTINCT ?n
WHERE{
?m a :Movie;
   :title ?t ;
   :genres ?g .
?g :name ?n .
}
```

Ensures unique results

QUERY RESULTS

Table | Raw Response

Showing 1 to 20 of 20 entries

	n
1	"Fantasy"
2	"Music"
3	"Drama"
4	"Comedy"
5	"History"
6	"Adventure"
7	"Action"
8	"Crime"
9	"Thriller"
10	"Documentary"
11	"Foreign"
12	"Romance"

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More SPARQL concepts:

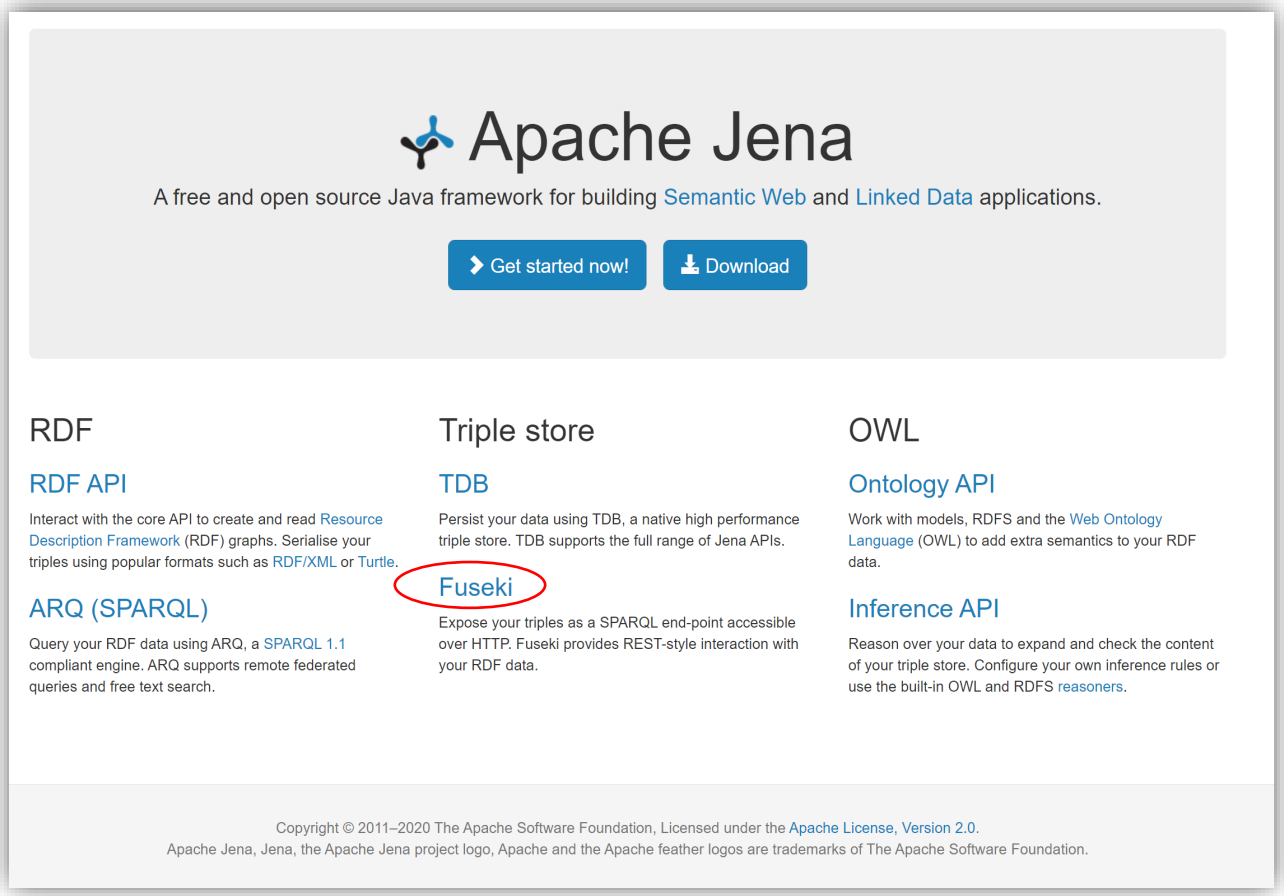
http://www.w3.org/TR/sparql11-query

- ASK queries: checks for a condition and returns a boolean result
- COUNT queries: count the number of results
- FILTER: allows to formulate additional conditions, e.g., on datatypes
- OPTIONAL: specifies optional values
- EXISTS / NOT EXISTS: negation
- GROUP BY, HAVING: aggregation
- ORDER BY: sorting
- Subqueries
- CONSTRUCT queries: generate new RDF statements like production rules
- …
- For details see http://www.w3.org/TR/sparql11-query

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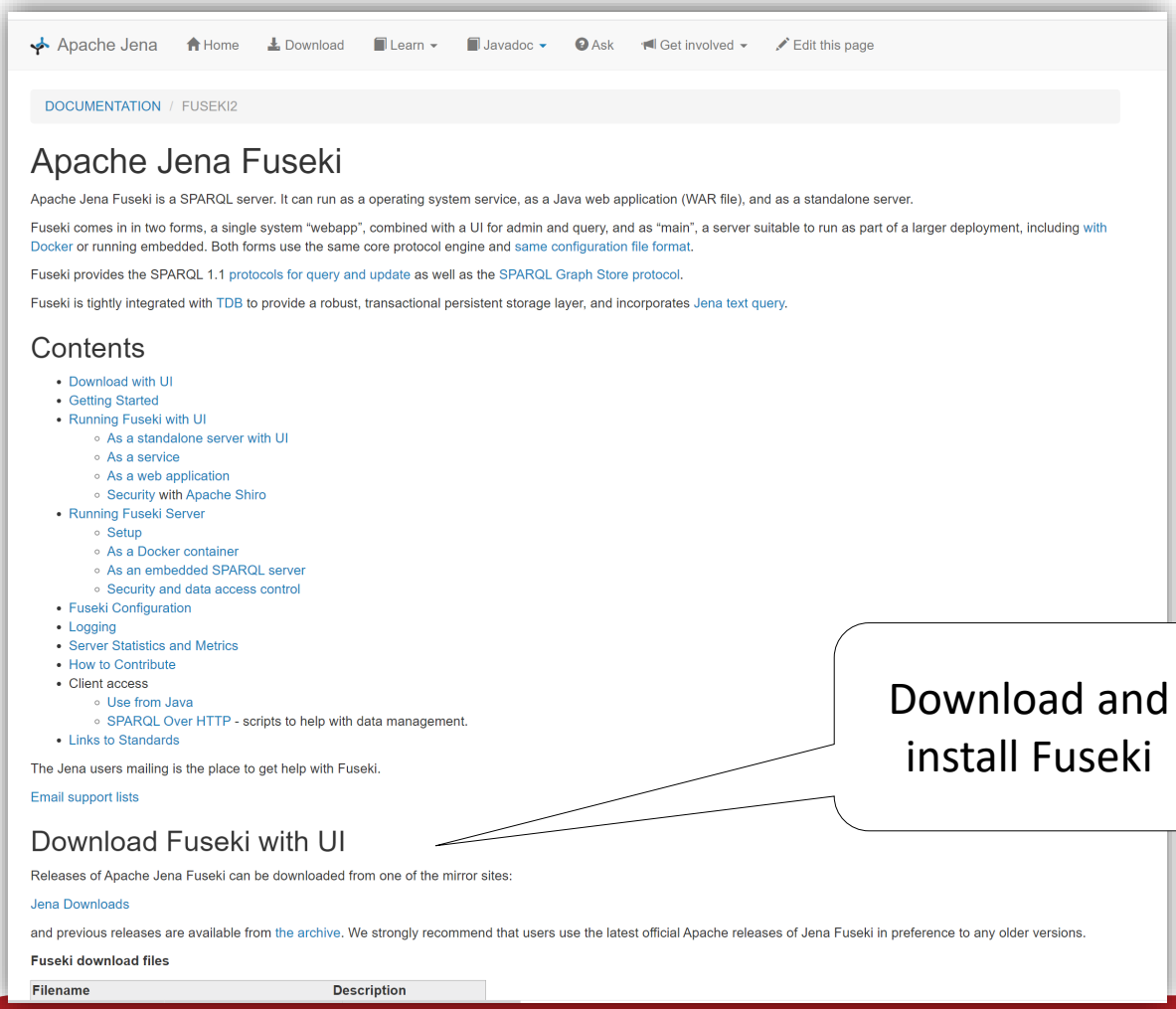
Apache Jena
Java framework for linked data



https://jena.apache.org

Prof. Dr. Hamed Shariat Yazdi
Department of Computer Science
Darmstadt Univeristy of Applied Sciences

Apache Jena Fuseki SPARQL server



Apache Jena | Home | Download | Learn | Javadoc | Ask | Get involved | Edit this page

DOCUMENTATION / FUSEKI2

Apache Jena Fuseki

Apache Jena Fuseki is a SPARQL server. It can run as a operating system service, as a Java web application (WAR file), and as a standalone server.

Fuseki comes in in two forms, a single system "webapp", combined with a UI for admin and query, and as "main", a server suitable to run as part of a larger deployment, including with Docker or running embedded. Both forms use the same core protocol engine and same configuration file format.

Fuseki provides the SPARQL 1.1 protocols for query and update as well as the SPARQL Graph Store protocol.

Fuseki is tightly integrated with TDB to provide a robust, transactional persistent storage layer, and incorporates Jena text query.

Contents

- Download with UI
- Getting Started
- Running Fuseki with UI
  - As a standalone server with UI
  - As a service
  - As a web application
  - Security with Apache Shiro
- Running Fuseki Server
  - Setup
  - As a Docker container
  - As an embedded SPARQL server
  - Security and data access control
- Fuseki Configuration
- Logging
- Server Statistics and Metrics
- How to Contribute
- Client access
  - Use from Java
  - SPARQL Over HTTP - scripts to help with data management.
- Links to Standards

The Jena users mailing is the place to get help with Fuseki.

Email support lists

Download Fuseki with UI

Releases of Apache Jena Fuseki can be downloaded from one of the mirror sites:

Jena Downloads

and previous releases are available from the archive. We strongly recommend that users use the latest official Apache releases of Jena Fuseki in preference to any older versions.

Fuseki download files

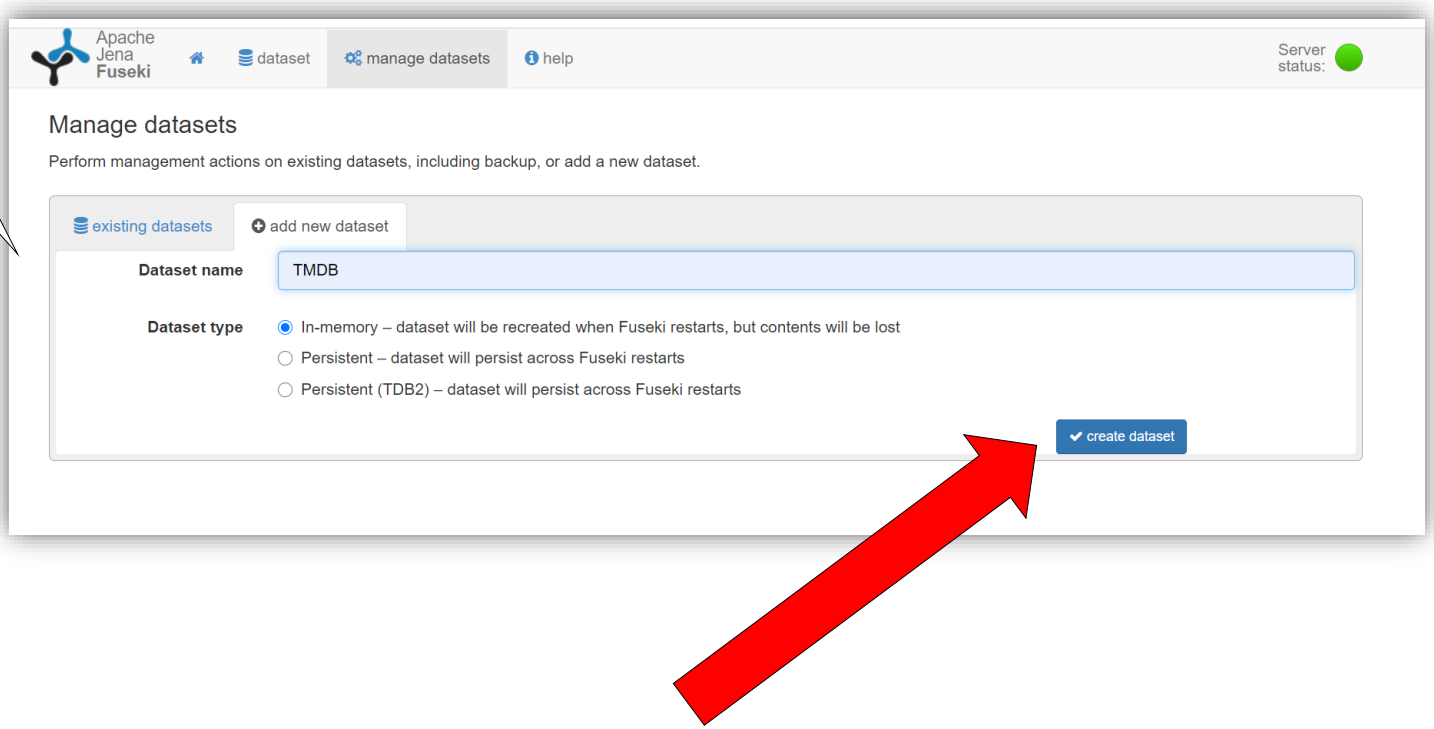
Filename	Description

Download and install Fuseki

https://jena.apache.org/documentation/fuseki2

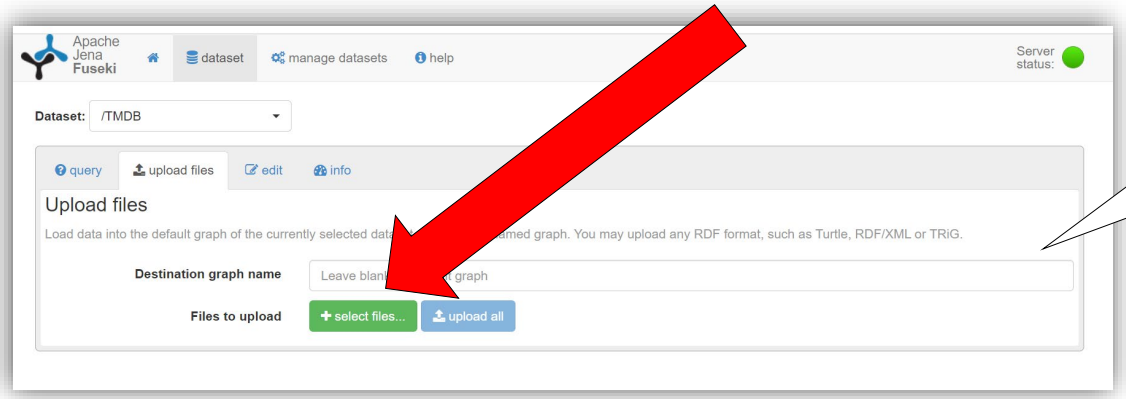
Start Fuseki server and web client
Add dataset TMDB

Add dataset TMDB

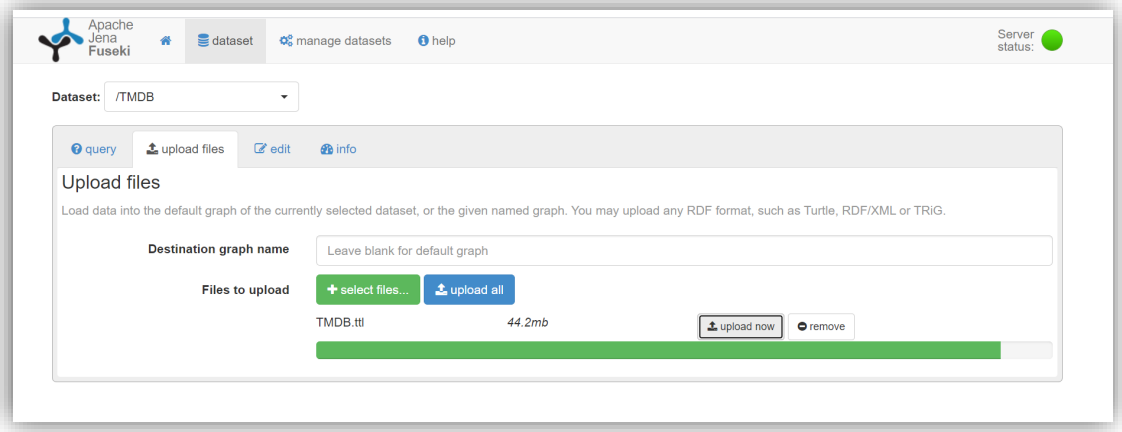


http://localhost:3030/manage.html

Upload knowledge graph TMDB.ttl



Upload TMDB.ttl



Write and execute SPARQL queries

```
1  PREFIX : <https://www.themoviedb.org/kaggle-export/>
2
3  SELECT ?m ?t ?g ?n
4  WHERE{
5  ?m a :Movie;
6     :title ?t ;
7     :genres ?g .
8  ?g :name ?n .
9  }
10
11
```

Execute SPARQL query

QUERY RESULTS

Table | Raw Response

Showing 1 to 50 of 7,511 entries

Search: Show 50 entries

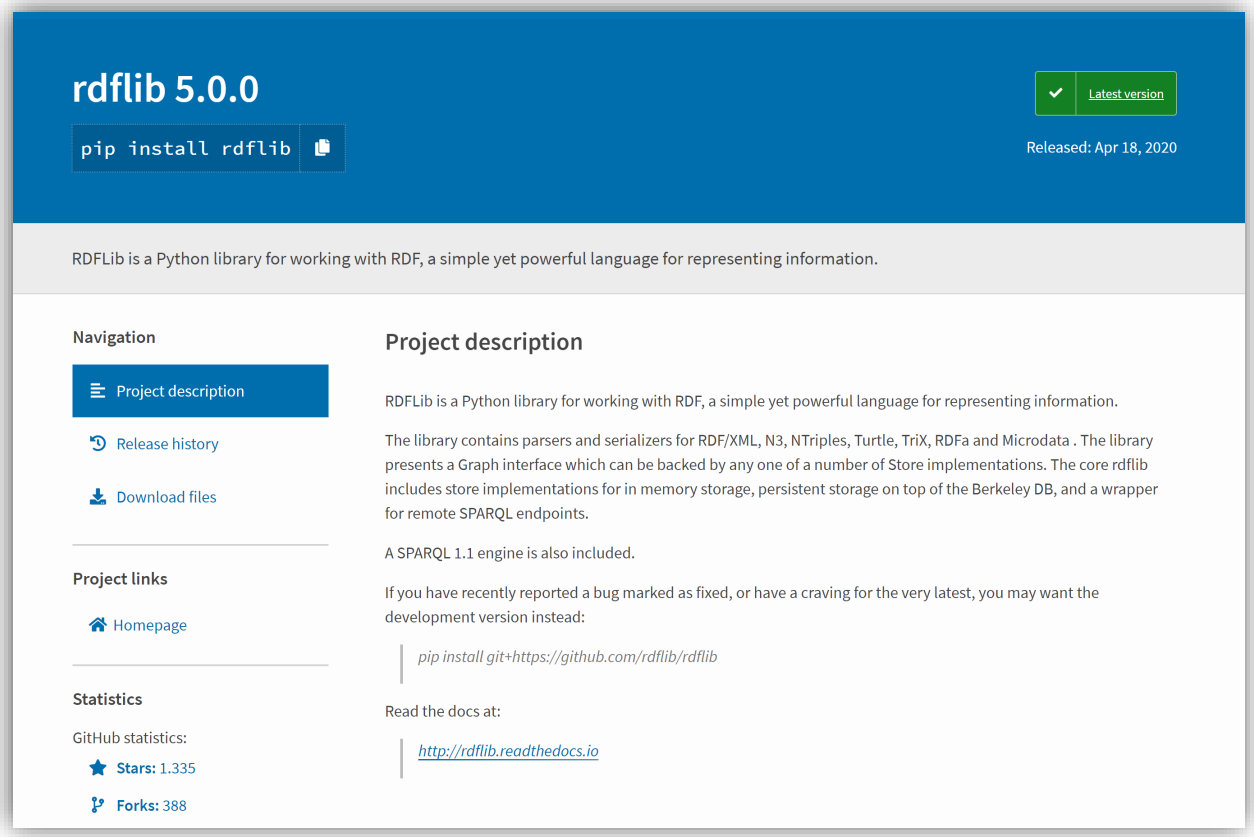
	m	t	g	n
1	:Movie_1622	"Cabin in the Sky"	:Genre_14	"Fantasy"
2	:Movie_1622	"Cabin in the Sky"	:Genre_10402	"Music"
3	:Movie_2191	"The Women"	:Genre_18	"Drama"
4	:Movie_2191	"The Women"	:Genre_35	"Comedy"
5	:Movie_235	"The Lost City of Z"	:Genre_18	"Drama"
6	:Movie_235	"The Lost City of Z"	:Genre_36	"History"
7	:Movie_235	"The Lost City of Z"	:Genre_12	"Adventure"
8	:Movie_235	"The Lost City of Z"	:Genre_28	"Action"
9	:Movie_1187	"Candy Razors"	:Genre_80	"Crime"
10	:Movie_1187	"Candy Razors"	:Genre_18	"Drama"
11	:Movie_2353	"Earthquake"	:Genre_53	"Thriller"

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- The TMDB Knowledge Graph
- SPARQL
- Apache Fuseki
- Using RDF and SPARQL in Python
- Quick check

rdflib: Python package for RDF/SPARQL

```
pip install rdflib
```



Loading a knowledge graph in rdflib

```
import rdflib

g = rdflib.Graph()
g.parse('data/TMDB.ttl', format='turtle')
```

Loading may take a while depending on the size of the knowledge graph

Executing SPARQL queries

```
query_result = g.query(
   """
   SELECT ?n
   WHERE{
      ?g a :Genre;
         :name ?n .
   }
   """)

for row in query_result:
    print(row)
```

Query result (type SPARQLResult) can be iterated: bindings with variables and values



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Quick check:
Knowledge representation

1. What is a knowledge graph?
2. Give examples for individuals, classes, relationships
3. What is a resource in RDF?
4. Which parts does an RDF triple consist of?
5. Name the classes of the TMDB knowledge graph used in this course
6. What is SPARQL?
7. How are simple queries expressed in SPARQL?
8. How many rows and columns does a query result have?
9. How to express conditions in queries?
10. How to join in SPARQL?
11. How to avoid duplicates in the result set?